1) Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in.

Practical: Install and Configure a DNS Server for musicmania.in

You can complete this networking lab using BIND9 on Ubuntu Linux or a Windows DNS application. Below are the steps for Ubuntu Linux using BIND9, which is commonly used to learn DNS server configuration.

1. Requirements

- A computer or virtual machine running Ubuntu Linux.
- Internet access to install BIND9.
- Terminal access with administrator (sudo) privileges.

The goal is to install the DNS server, create a DNS zone named musicmania.in, configure DNS records, and test name resolution locally.

2) Add an A record in your DNS server configuration to map www.musicmania.in to the IP address 192.168.1.50, then use the nslookup or dig command to verify the mapping works.

Step 1: Add the A record

On your Ubuntu machine, open the BIND9 zone file:

sudo nano /etc/bind/db.musicmania.in

Add or update this record:

www IN A 192.168.1.50

This maps www.musicmania.in to the IP address 192.168.1.50.

Step 2: Validate and restart DNS

Run these commands:

sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in

sudo systemctl restart bind9

The zone check should return OK.

Step 3: Verify using nslookup

Query your local DNS server:

nslookup www.musicmania.in 127.0.0.1

Expected output:

Server: 127.0.0.1

Address: 127.0.0.1#53

Name: www.musicmania.in

Address: 192.168.1.50

Alternatively, use dig:

dig @127.0.0.1 www.musicmania.in A

Look for the answer:

;; ANSWER SECTION:

www.musicmania.in. 86400 IN A 192.168.1.50

3) Simulate a typo by adding a CNAME record for play.musicmania.in that points to www.musicmania.in, then test if accessing play.musicmania.in resolves to the correct IP address.

play.musicmania.in is an alias of www.musicmania.in.

www.musicmania.in resolves to 192.168.1.50.

Therefore, play.musicmania.in should also resolve to 192.168.1.50.

**4) Your DNS server is not resolving the domain musicmania.in as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with nslookup or dig.

Hint: Include at least three different troubleshooting commands or checks in your documentation.**

Problem	Possible cause	Solution
DNS service is stopped	BIND9 service failure	Restart BIND9
Configuration check fails	Syntax error	Correct named.conf.local
Zone check fails	Incorrect zone file	Fix DNS records and SOA syntax
NXDOMAIN response	Record is missing or hostname is incorrect	Add or correct the record
1`Wrong IP address	Incorrect or outdated A record	Update the record and increment the SOA serial
Query times out	DNS service, network, or firewall issue	Check port 53 and service logs